

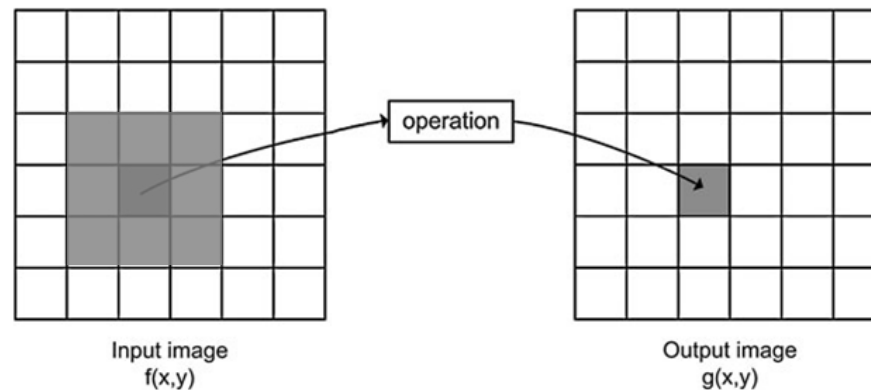
Biomedical Image Processing

FILTERING IN THE SPATIAL DOMAIN

Helena R. Torres

Local operations

- ❑ Unlike point operations, local operations use the neighborhood of a pixel to estimate the new intensity for that pixel;
- ❑ In these operations, each point of the resulting image $g(i,j)$ is obtained from a T transformation of the homologous coordinate point of the initial image $f(i,j)$ and its neighborhood;
- ❑ Local Operations -> Operations



Spatial filtering

□ Process an image so that the resulting image is more appropriate than the original image for a specific application

□ Function expressed as:

$$g(x, y) = T[f(x, y)]$$

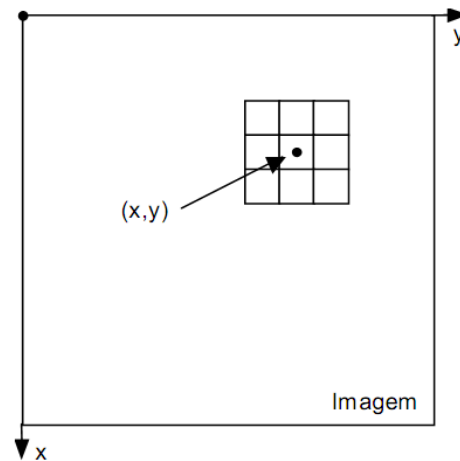
- $g(x, y)$ is the processed image
- $f(x, y)$ is the original image
- T is an operator in f , defined in a certain neighbourhood of (x, y)

□ In practice, filtering an image is done through kernel convolution operations, so the expression before is written as:

$$g = f * k$$

Spatial filtering

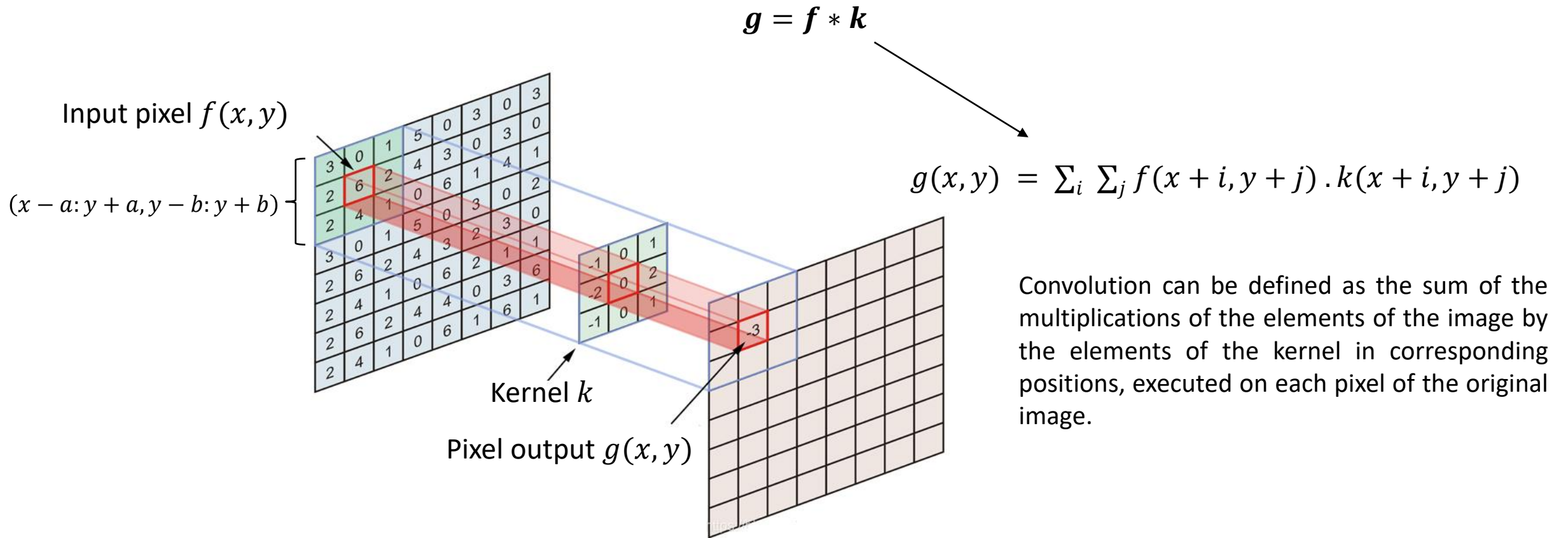
- ❑ Neighborhood around (x, y)
- ❑ Example: 8-pixel neighborhood equals to a 3x3 kernel



- ❑ The kernel is moved pixel by pixel along $f(x,y)$ and each pixel is applied to the T operator to calculate the value of $g(x,y)$ at that point.



Spatial filtering

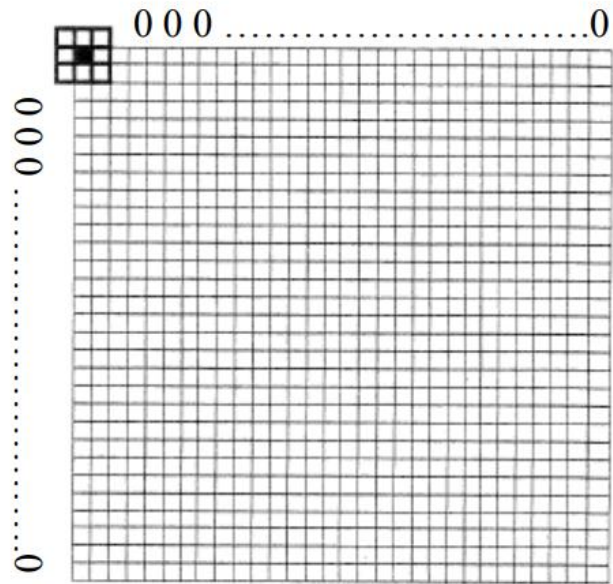


Spatial filtering

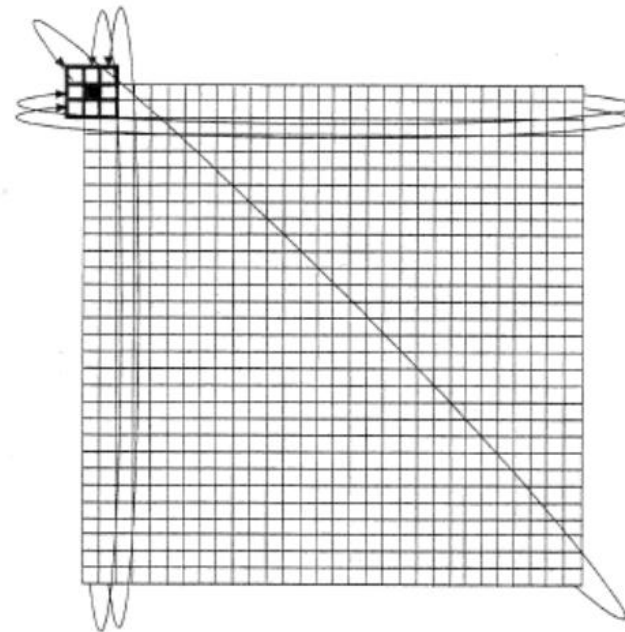
- Treatment of pixels on the borders



Zero padding



ou



Spatial filtering

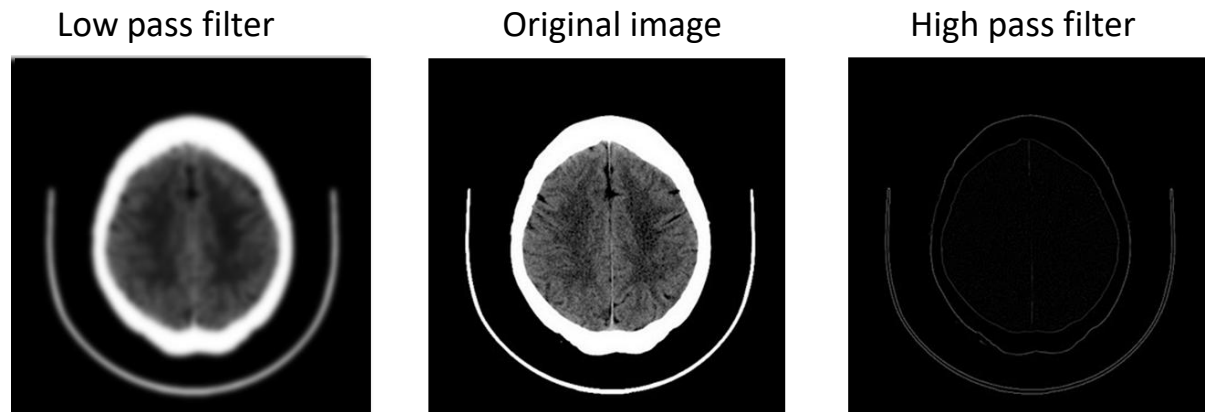
PSEUDOCODE TO FILTER THROUGH KERNELS

- 1) Apply padding to the original image
- 2) Create image of zeros that will contain the transformed image
- 3) Scroll through the image with padding
- 4) Compute the neighborhood with the region of interest to use to filter the image
 - 1) Multiply by the chosen kernel
 - 2) Sum all result values and place in the corresponding pixel in the transformed image

Spatial filtering

- ❑ A spatial filter is composed of:
 - ❑ A neighborhood
 - ❑ An operation that is carried out on the points of a neighborhood
- ❑ The operation is usually performed through convolution with specific kernels

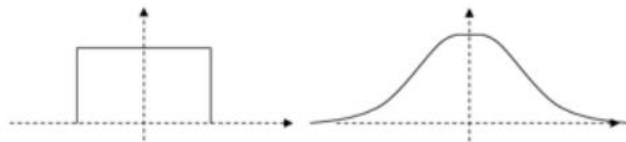
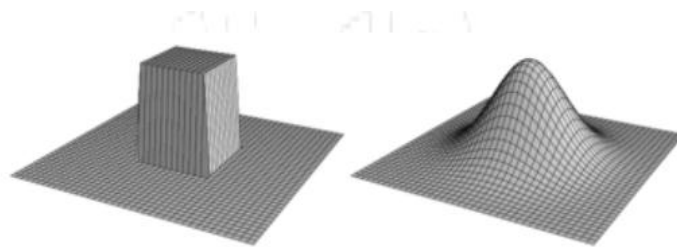
- ❑ Filters can be classified into:
 - ❑ Low pass - noise removal
 - ❑ High pass - contour detection
 - ❑ Bandpass



Spatial filtering

FILTER TYPES

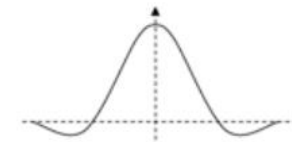
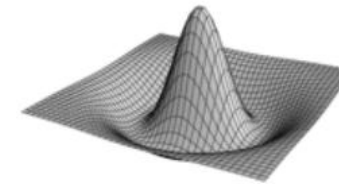
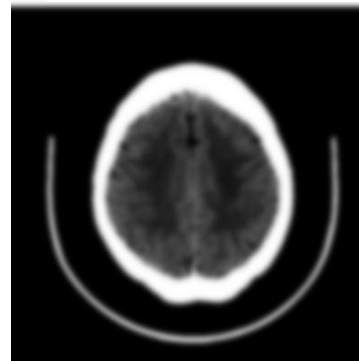
- Depending on the filter used (kernel), the operations to be performed on the image are different



0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

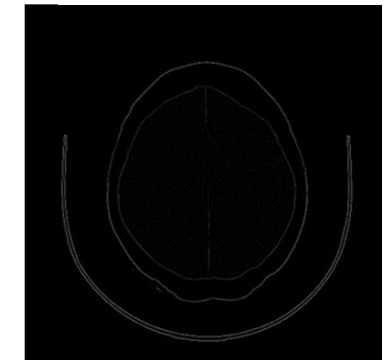
0	1	2	1	0
1	3	5	3	1
2	5	9	5	2
1	3	5	3	1
0	1	2	1	0

Low pass filter



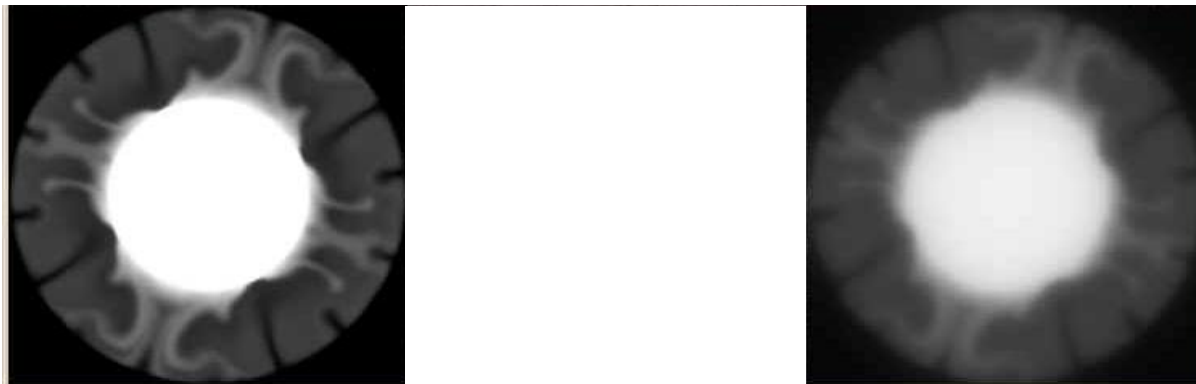
0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

High pass filter



Low Pass Filters

- ❑ Attenuate or eliminate high-frequency components
 - noise, areas of significant changes in intensity
- ❑ Image smoothing
- ❑ Noise reduction and small details
- ❑ The weights (values) of kernels should be positive and their sum should be 1



Noise

- ❑ Noise is a disruption to image data by the image acquisition system or by the transmission of images;
- ❑ There are several types of noise: additive (e.g. Gaussian), multiplicative, impulsive (e.g. salt-and-pepper)



gaussian



salt-and-pepper

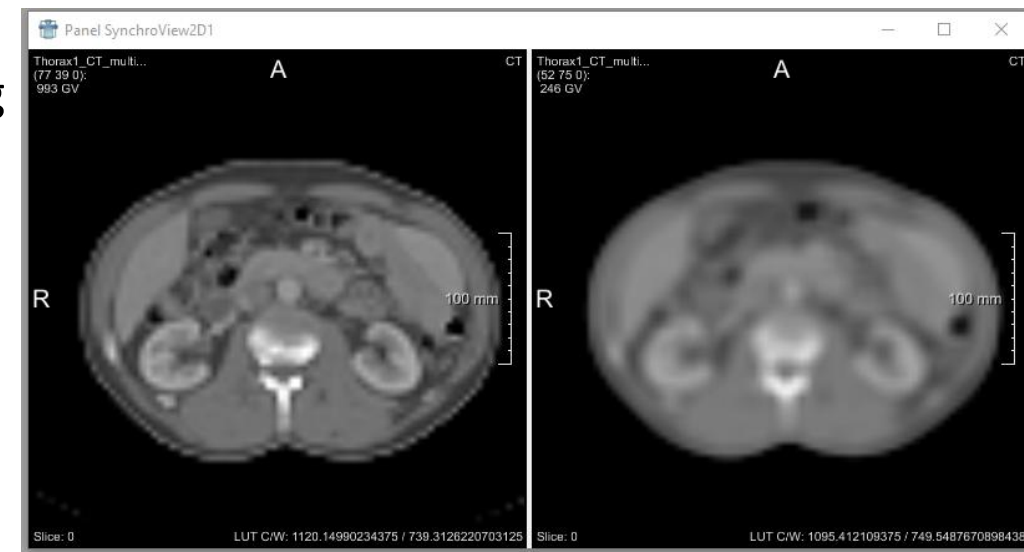
Low Pass Filters

AVERAGE FILTER

- ❑ Smooths the image from the average intensity of the pixels in the vicinity
- ❑ Allows you to reduce noise
- ❑ No preservation of edges
- ❑ The larger the neighborhood, the greater the smoothing
- ❑ You can also use kernel convolution:

- *kernel 3x3*

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

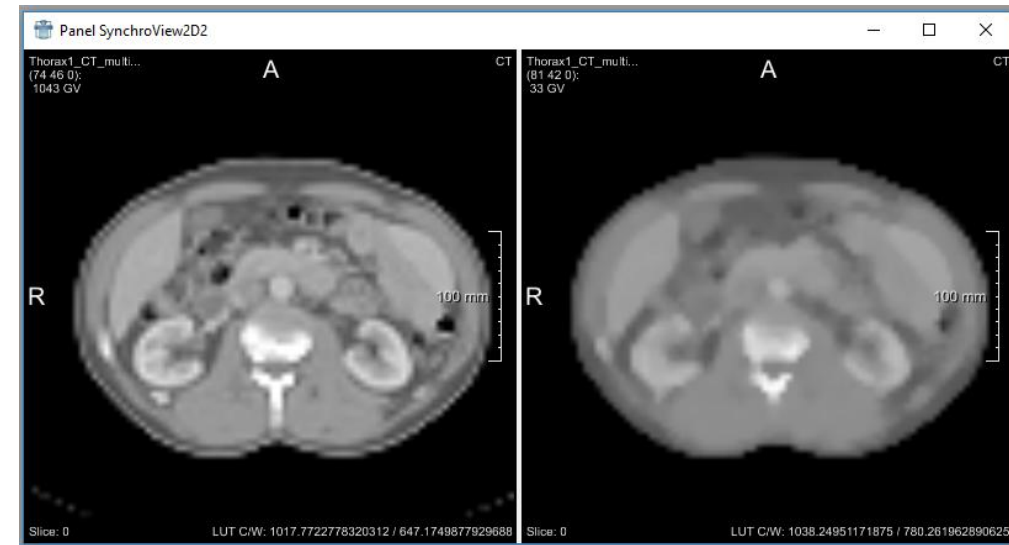
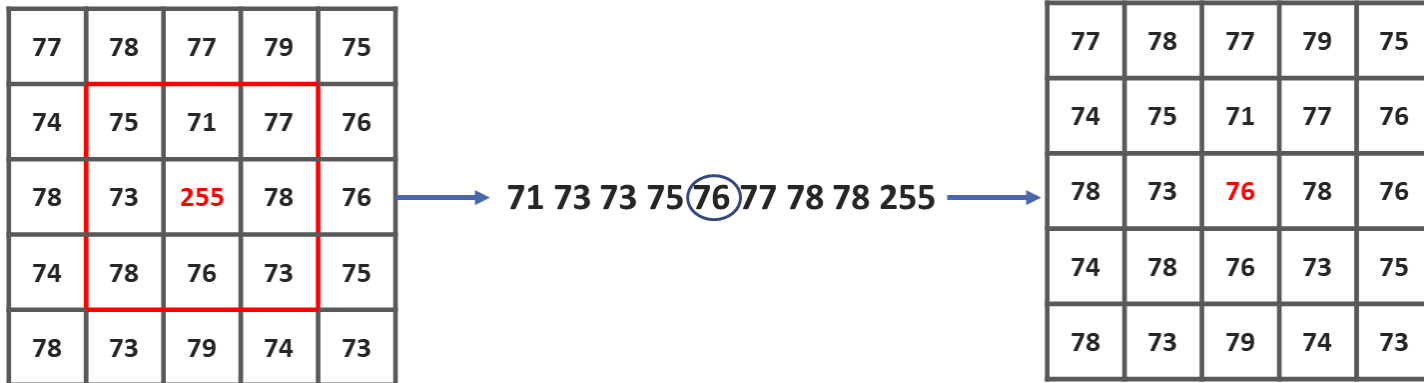


Low Pass Filters

MEDIAN FILTER

- ❑ Smooths the image from the median intensity of pixels in the vicinity
- ❑ Removes noise by eliminating high/low-intensity pixels without oversmoothing
 - Eliminates impulsive noise (salt-and-pepper)

❑ Example:



Low Pass Filters

GAUSSIAN FILTER

☐ SMOOTHES THE IMAGE USING A CONVOLUTION OPERATOR WHOSE VALUES REPRESENT A GAUSSIAN DISTRIBUTION

☐ Kernel values are set by:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

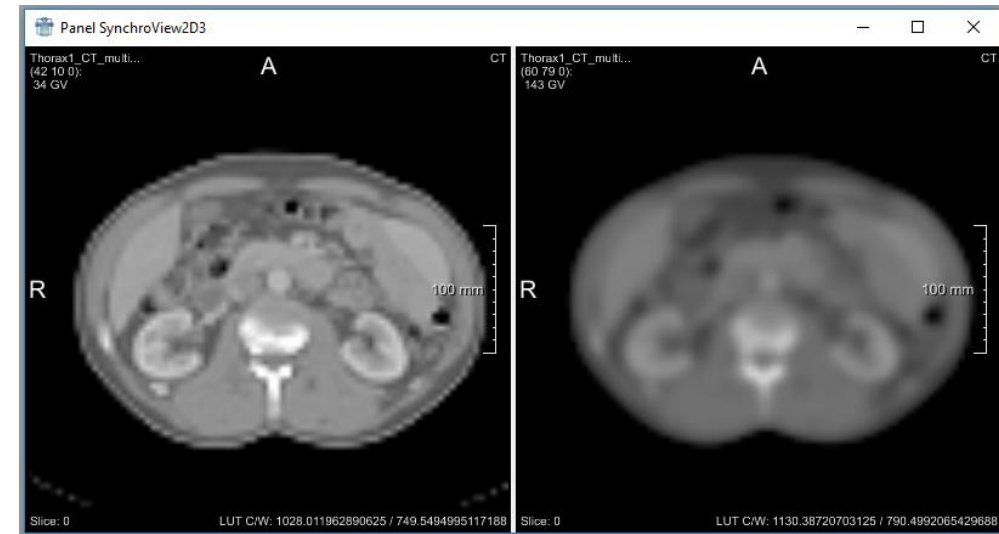
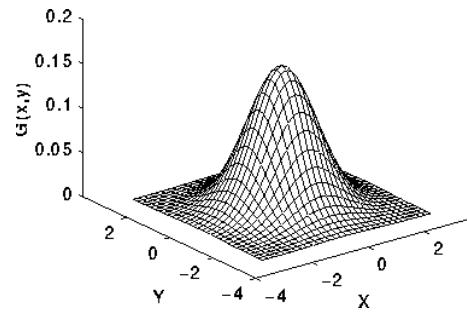
- σ controls the degree of smoothing, the higher σ the higher the smoothing

☐ One of the most widely used filters in medical imaging

☐ Exemplo:

- *kernel 3x3*

1/16	2/16	1/16
2/16	4/16	2/16
1/16	2/16	1/16



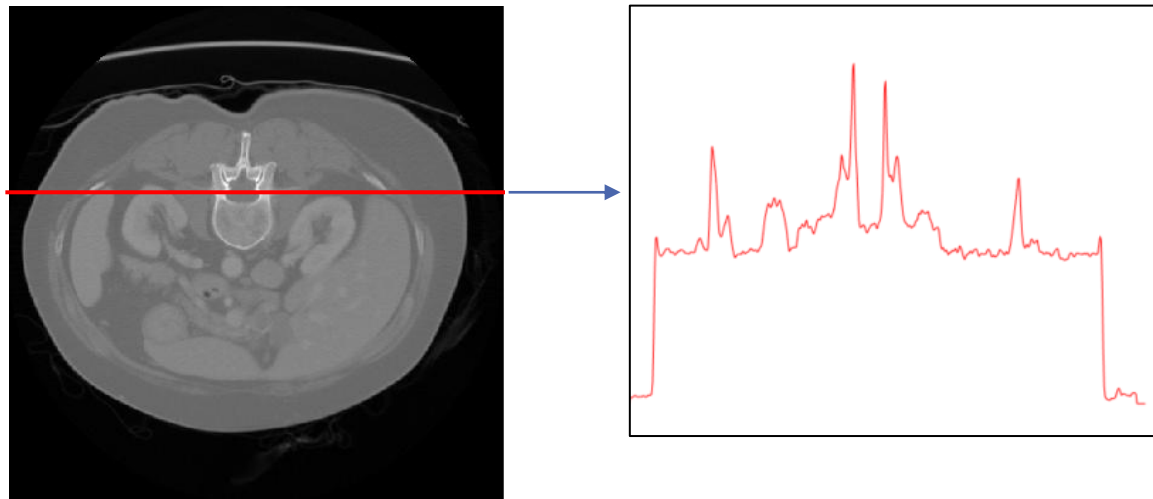
High Pass Filters

- ❑ Emphasize high-frequency components
 - areas of significant changes in intensity
- ❑ Enhance fine image details (sharpening)
- ❑ Contour design
- ❑ The sum of kernel coefficients should be zero.
 - ❑ As soon as applied to a homogeneous region of the image the result should be close to zero

High Pass Filters

CONTOUR DETECTION

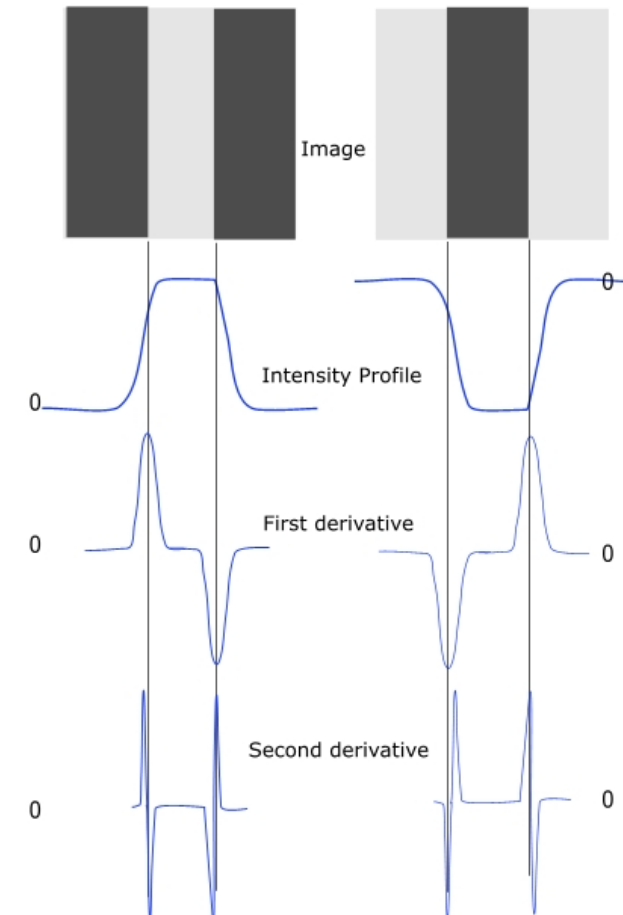
- ❑ Contours: Area of the image where there is a significant variation in intensity
- ❑ Allows you to extract information from the location of objects, their shape, their size, ...



High Pass Filters

CONTOUR DETECTION

- Contour scan can be done through the image gradient:
 - Detection of the local maximum or minimum location of the first derivative.
 - Detection of the "passages through zero" of the second derivative.
- More precisely, the gradient is a vector that stores information from the partial derivatives of the image (derived in x and y)



High Pass Filters

GRADIENT

- In image, the **grad(f)** gradient can be divided into:

$$\frac{\partial f}{\partial x} = f(i, j+1) - f(i, j)$$

$$\frac{\partial f}{\partial y} = f(i, j) - f(i+1, j)$$



Using kernels:

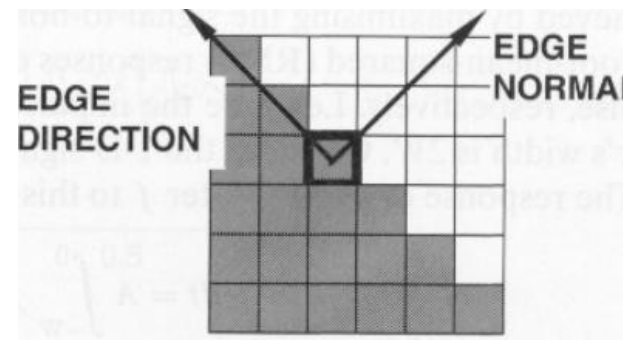
$$\begin{bmatrix} -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

- Because it is a vector, the gradient has a module and direction

$$magnitude(grad(f)) = \sqrt{\frac{\partial f^2}{\partial x} + \frac{\partial f^2}{\partial y}}$$

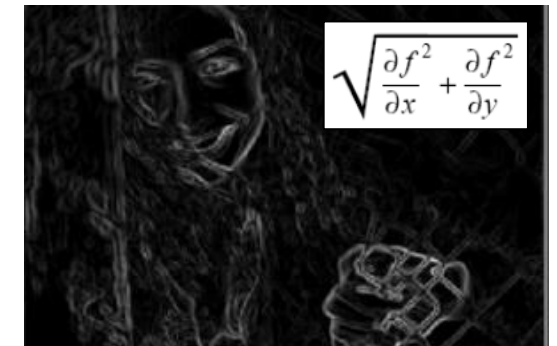
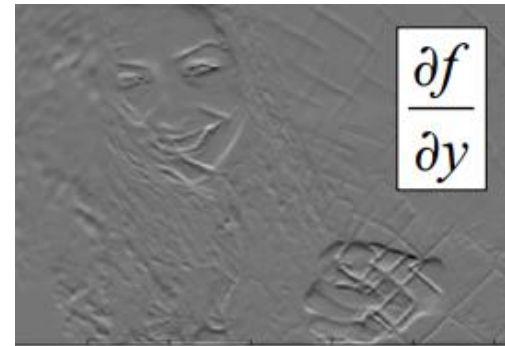
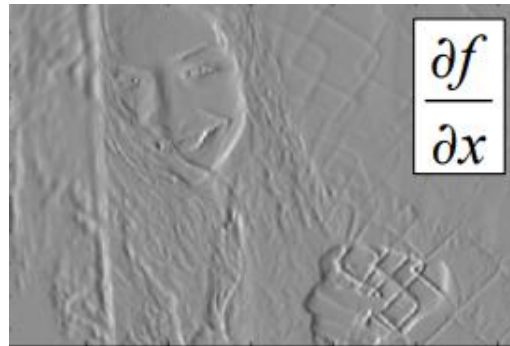
$$direction(grad(f)) = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$



Module: transition "strength"
Direction: Direction of transition

High Pass Filters

GRADIENT



- ❑ In order to facilitate the perception of contours or image enhancement, you can filter the image with derivative filters.
- ❑ This is equivalent to convolving the image with kernels that allow you to obtain the information of the gradients

High Pass Filters

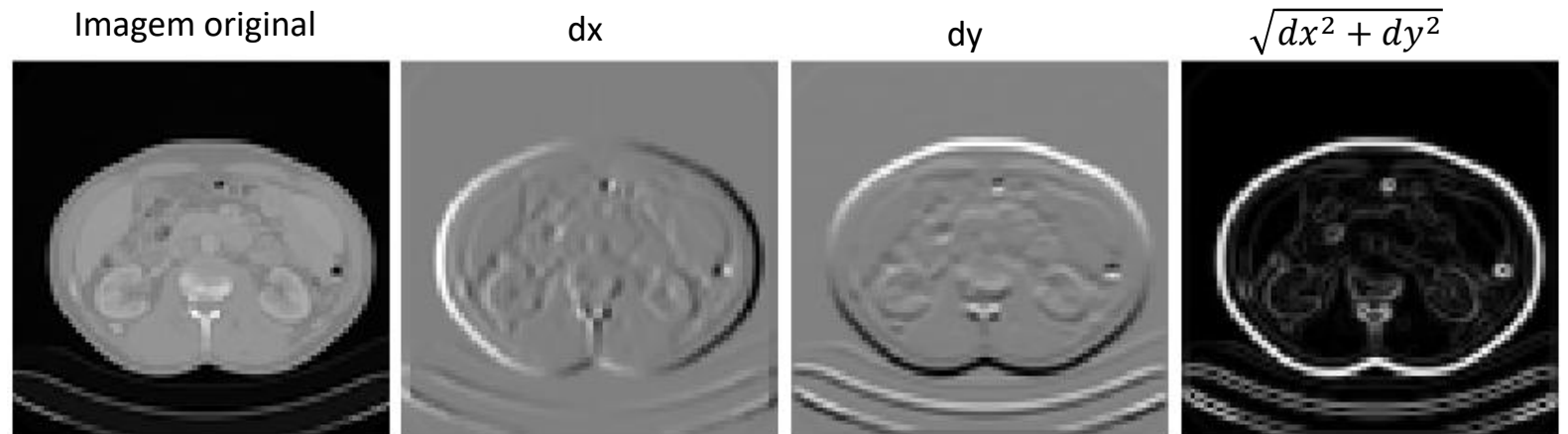
SOBEL FILTER

- ❑ Calculates the image gradient in the X or Y direction
 - First derivative in its direction
- ❑ Contour detection
- ❑ Often preceded by a low-pass filter to eliminate noise
- ❑ *Kernels:*

1	0	-1
2	0	-2
1	0	-1

1	2	1
0	0	0
-1	-2	-1

Horizontal direction (dx) Vertical direction (dy)



High Pass Filters

LAPLACE FILTER

- ❑ Calculates the second derivative of the image
- ❑ Highlight rapid intensity transitions
- ❑ Contour detection
- ❑ Often preceded by a low-pass filter
- ❑ *Kernel:*

0	1	0
1	-4	1
0	1	0

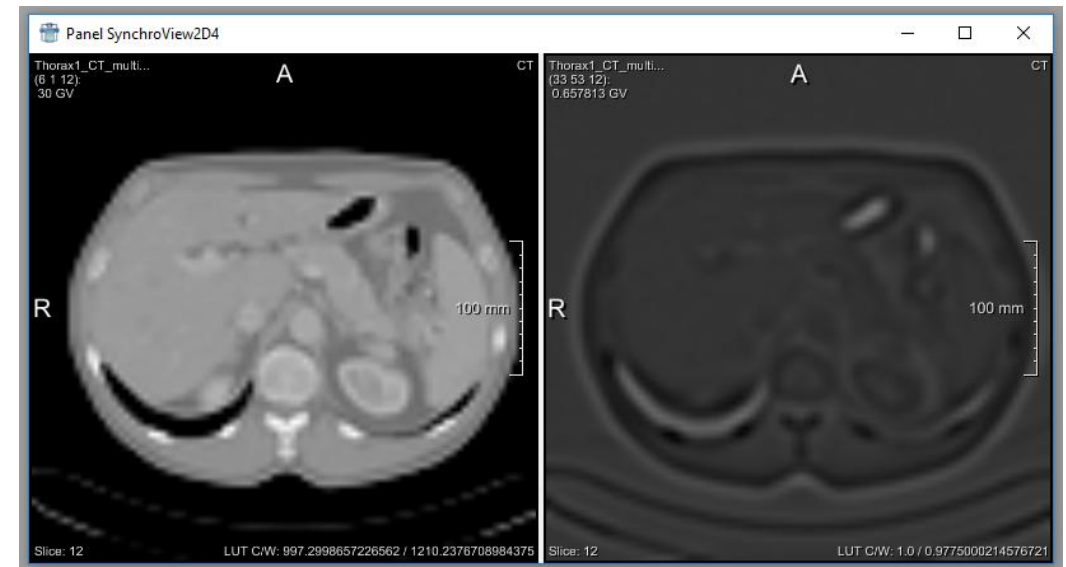
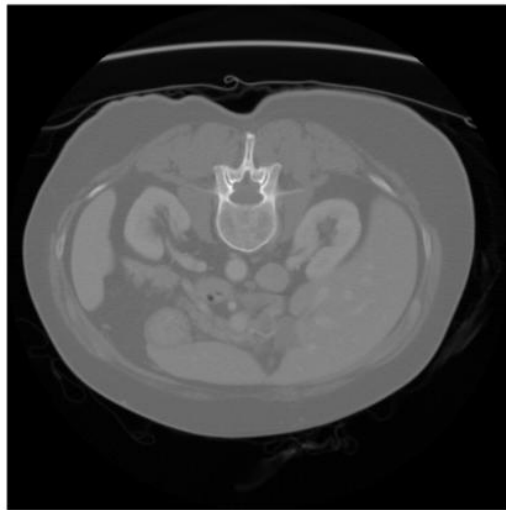


Image enhancement

SHARPENING - HIGHBOOST

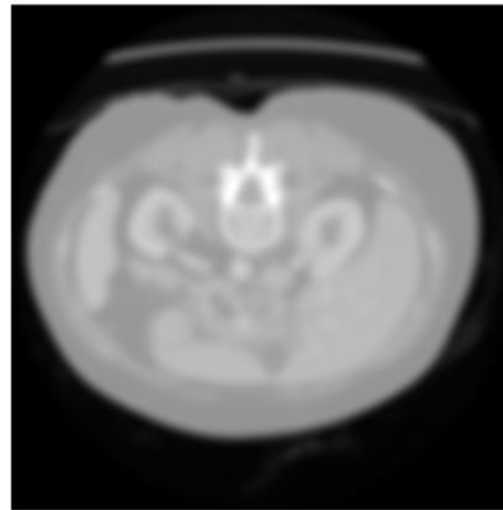
- Highlighting the outlines of an image can also be achieved by subtracting a smoothed image (low-pass filter) from the original image

$$I_{\text{sharp}} = I_{\text{original}} - I_{\text{smooth}}$$



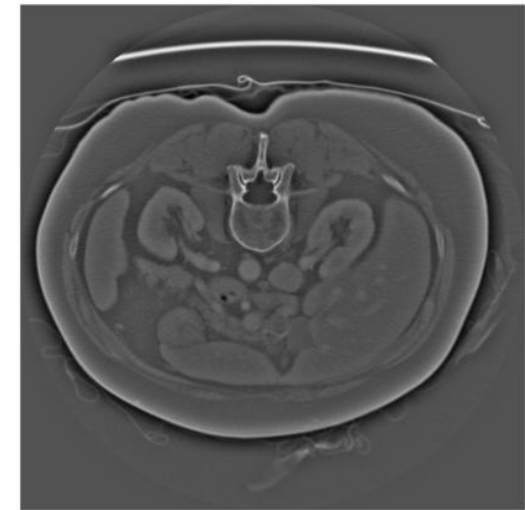
Original image - I_{original}

-



Smoothed image - I_{smooth}

=



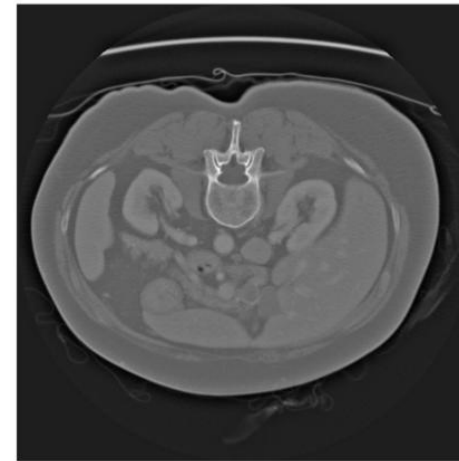
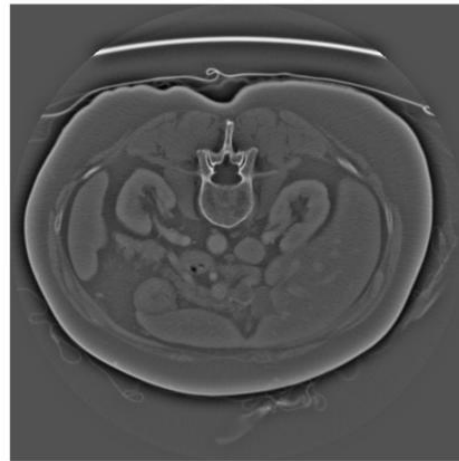
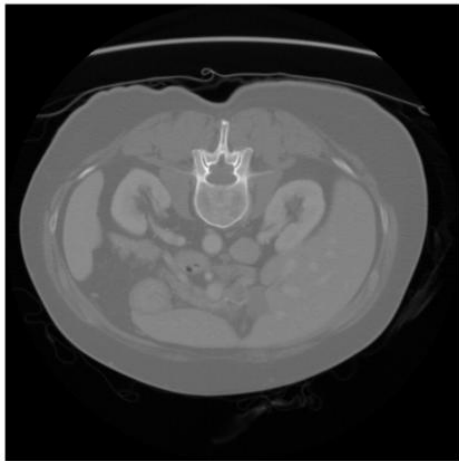
Final image - I_{sharp}

Image enhancement

SHARPENING - HIGHBOOST

- However, contour highlighting emphasizes transitions but image details are lost (low frequency components)
- The highboost filter amplifies the input image (by a factor A) and then subtracts by a smoothed version of the image

$$I_{\text{highboost}} = A I_{\text{original}} - I_{\text{smooth}} = (A-1) I_{\text{original}} + I_{\text{original}} - I_{\text{smooth}} = (A-1) I_{\text{original}} + I_{\text{sharp}}$$



Equivalent to convolving the image with a kernel:

-1	-1	-1
-1	w	-1
-1	-1	-1

-1	-1	-1
-1	17	-1
-1	-1	-1

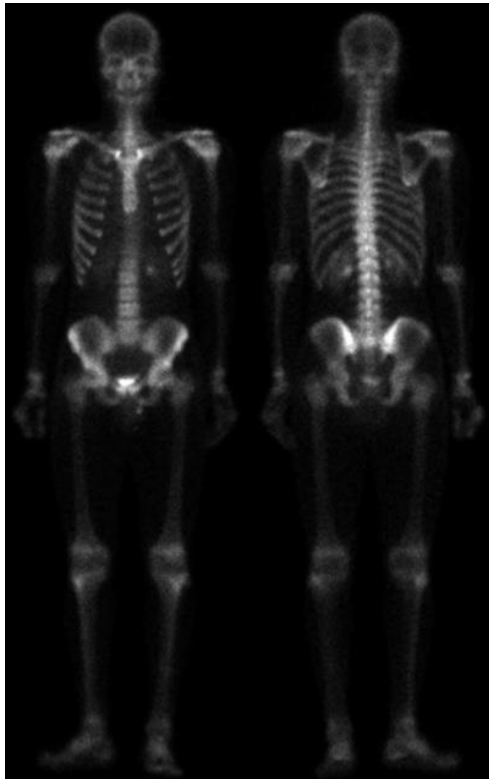
$$W = 9A - 1$$

- $A = 1$: High pass
- $A > 1$: The original image is added to the high pass

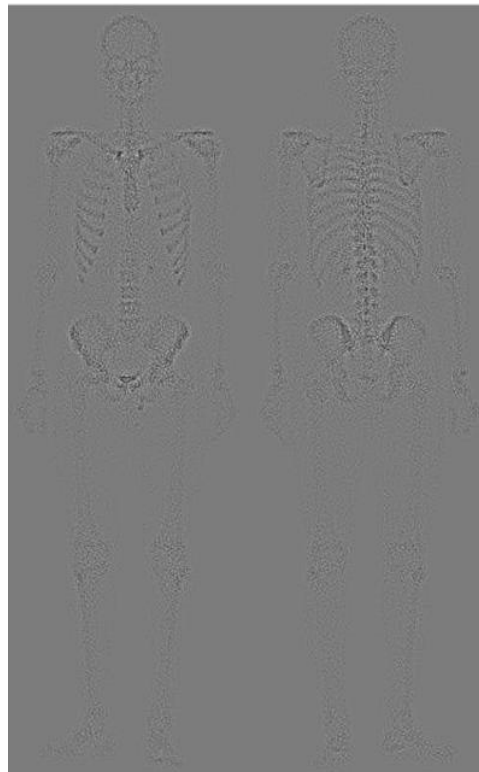
Image enhancement

FILTER COMBINATION - ENHANCEMENT ALGORITHM

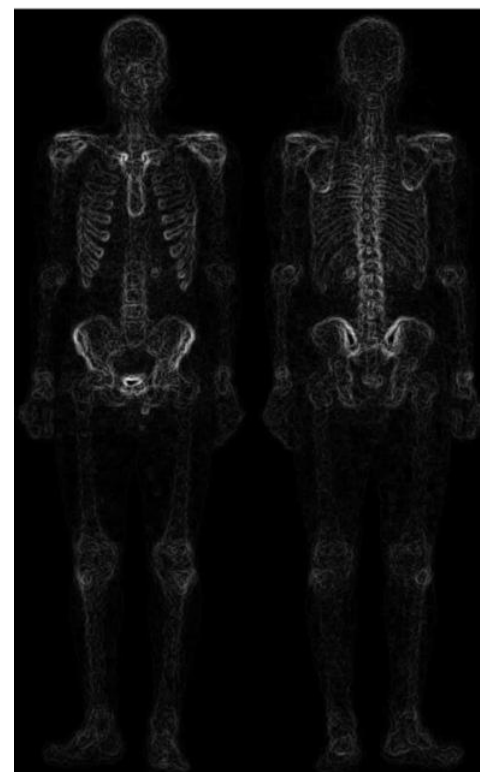
I_{original}



$I_{\text{lap}} : \text{laplacian of } I_{\text{original}}$



$I_{\text{sobel}} : \text{sobel of } I_{\text{original}}$



$I_{\text{real}} : I_{\text{lap}} + I_{\text{original}}$

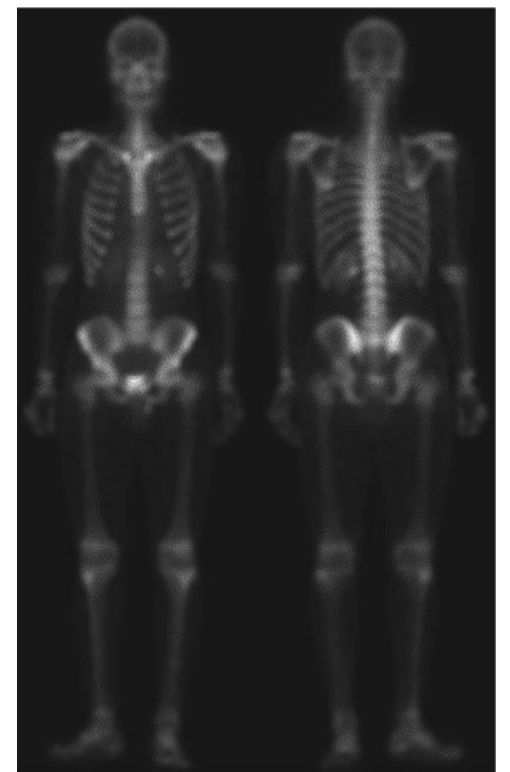
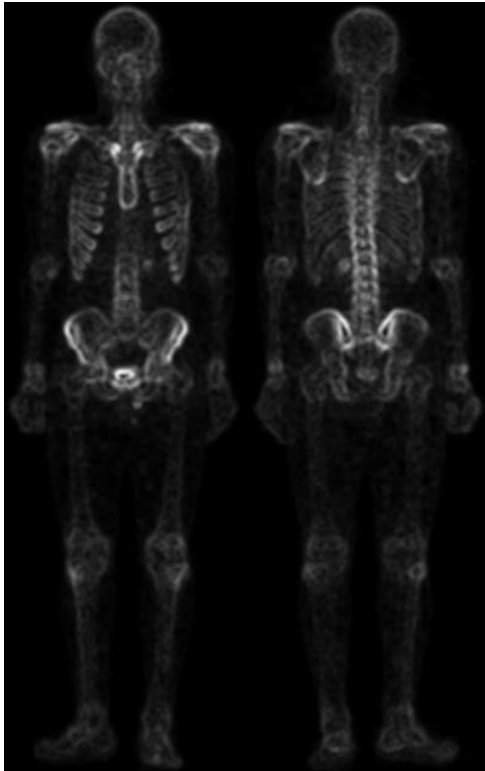


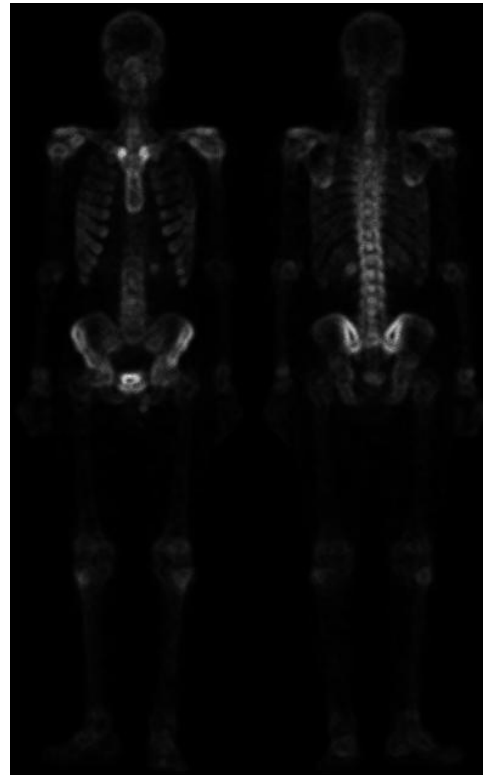
Image enhancement

FILTER COMBINATION - ENHANCEMENT ALGORITHM

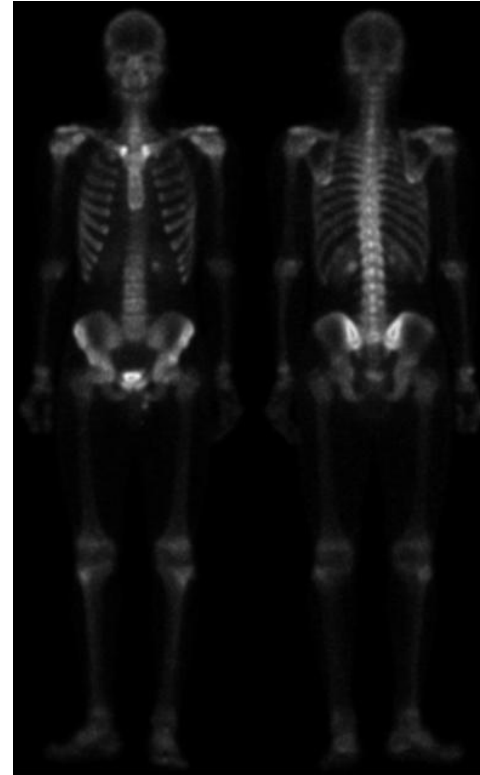
I_{smooth} : average of I_{sobel} (3x3)



I_{prof} : $I_{\text{real}} \times I_{\text{smooth}}$



I_{sharp} : $I_{\text{original}} + I_{\text{prof}}$



I_{final} : I_{sharp} Equalized

